

The Simplified CDP Model

Full Equation Reference

A single-factor (labor-only) version of Caliendo, Dvorkin & Parro (2019), “Trade and Labor Market Dynamics: General Equilibrium Analysis of the China Trade Shock,” Econometrica 87(3).

This document collects every equation of the simplified model in one place, alongside the corresponding equation number from the original paper. Simplification #1 (agreed so far): **no structures, no materials/input-output linkages, no rentiers/portfolio**. Firms in each region-sector market demand only labor. Everything else — the household’s dynamic migration problem — is untouched from the original paper.

1 Notation

Symbol	Ranges over	Meaning
n, i, m	$1, \dots, N$	regions
j, k, h	$0, 1, \dots, J$	sectors, where $j = 0$ denotes non-employment in that region
t	$0, 1, 2, \dots$	time

A “market” is a region-sector pair (n, j) . Sector 0 is a degenerate market in each region with fixed consumption b^n and no production, no wage, no trade — it exists purely so non-employment is one more option in the household’s discrete choice.

2 Fundamentals and parameters

Time-varying fundamentals $\Theta_t = (A_t, \kappa_t)$ — the two things a counterfactual (e.g. a productivity shock) perturbs:

- A_t^{nj} — productivity in region n , sector j (for $j = 1, \dots, J$; undefined for $j = 0$)
- $\kappa_t^{nj,ij}$ — iceberg trade cost shipping sector- j goods from region i to region n

Constant fundamentals $\bar{\Theta} = (\Upsilon, b)$ — fixed for all t . (Note: this is the original paper’s $\bar{\Theta} = (\Upsilon, H, b)$ with the structures endowment H dropped, since there is no structures factor left.)

- $\Upsilon = \{\tau^{nj,ik}\}$ — labor relocation (mobility) costs, in utility terms, from market (n, j) to market (i, k)
- b^n — value of home production in region n (non-employed household’s consumption)

Structural parameters — constant, but governing preferences/technology curvature rather than being region- or time-specific levels:

- $\beta \in [0, 1)$ — discount factor
- θ^j — trade elasticity (Fréchet shape parameter) in sector j

- ν — governs the migration elasticity ($1/\nu$ is the migration elasticity)
- α^j — Cobb-Douglas final-consumption expenditure share on sector j , with $\sum_{j=1}^J \alpha^j = 1$
- η^{nj} — CES elasticity of substitution across varieties within sector j in region n ; enters only through the constant Γ^{nj} in the sectoral price index (§4.3), subject to $1 + \theta^j > \eta^{nj}$

State variable: $L_t = \{L_t^{nj}\}_{n=1,\dots,N}^{j=0,\dots,J}$, the mass of households in each market at time t . This is the only object that carries information from one period to the next.

3 Household block (dynamic) — unchanged from the original paper

3.1 Preferences

For an employed household in market (n, j) , $j \geq 1$:

$$C_t^{nj} = \prod_{k=1}^J \left(c_t^{nj,k} \right)^{\alpha^k} \quad (\text{orig. eq. 1}) \quad (1)$$

a Cobb-Douglas bundle over the J sector final goods, **all purchased locally in region n** — the household never chooses which region to buy from; that choice already happened upstream (Section 4). For a non-employed household, $C_t^{n0} = b^n$.

The household's ideal price index (region-specific, not sector-specific — the same for every employed household in region n regardless of which sector j they work in):

$$P_t^n = \prod_{k=1}^J \left(\frac{P_t^{nk}}{\alpha^k} \right)^{\alpha^k}$$

so that real consumption for an employed household is $C_t^{nj} = w_t^{nj}/P_t^n$, and utility is logarithmic (orig. Assumption 3): $U(C_t^{nj}) = \log(C_t^{nj})$.

3.2 Bellman equation and expected lifetime value

Idiosyncratic shocks are i.i.d. Type-I Extreme Value (orig. Assumption 2). After integrating them out, the expected lifetime utility of a representative household in market (n, j) is:

$$V_t^{nj} = U(C_t^{nj}) + \nu \log \left(\sum_{i=1}^N \sum_{k=0}^J \exp \left(\beta V_{t+1}^{ik} - \tau^{nj,ik} \right)^{1/\nu} \right) \quad (\text{orig. eq. 2}) \quad (2)$$

3.3 Migration shares

The share of households relocating from market (n, j) to market (i, k) :

$$\mu_t^{nj,ik} = \frac{\exp \left(\beta V_{t+1}^{ik} - \tau^{nj,ik} \right)^{1/\nu}}{\sum_{m=1}^N \sum_{h=0}^J \exp \left(\beta V_{t+1}^{mh} - \tau^{nj,mh} \right)^{1/\nu}} \quad (\text{orig. eq. 3}) \quad (3)$$

Note that the log-sum-exp term inside V_t^{nj} is exactly the (unnormalized) denominator of μ_t^{nj} . — V_t and μ_t are two views of the same object, computed from the same βV_{t+1} terms.

3.4 Law of motion for the state

$$L_{t+1}^{nj} = \sum_{i=1}^N \sum_{k=0}^J \mu_t^{ik,nj} L_t^{ik} \text{ (orig. eq. 4)} \quad (4)$$

4 Static / production block — simplified (labor only, no materials, no structures, no rentiers)

Applies only to real sectors, $j = 1, \dots, J$ (sector 0 has no production side at all).

4.1 Production

A firm producing a variety with efficiency draw z in market (n, j) :

$$q_t^{nj}(z) = z \cdot A_t^{nj} \cdot l_t^{nj}$$

Constant returns in labor alone — no value-added share γ^{nj} , no structures share ξ^n , no materials. Unit cost of producing one unit of the variety: $w_t^{nj} / (z A_t^{nj})$.

4.2 Price of a variety, sourced from the cheapest region

$$p_t^{nj}(z^j) = \min_i \left\{ \frac{\kappa_t^{nj,ij} w_t^i}{z^{ij} A_t^{ij}} \right\}$$

This is where trade actually happens — competition across all N regions for who supplies a given variety to region n 's sectoral aggregate good. It is unaffected by the simplification; the only thing that changed is what “unit cost” x_t^{ij} collapses to (just w_t^i , instead of the Cobb-Douglas cost-of-materials bundle in the full model).

4.3 Sectoral price index

Given the Fréchet distribution of productivity draws (orig. eq. 6, with the value-added share γ^{ij} set to 1 since there are no materials diluting local TFP's contribution):

$$P_t^{nj} = \Gamma^{nj} \left(\sum_{i=1}^N \left(w_t^i \kappa_t^{nj,ij} \right)^{-\theta^j} \left(A_t^{ij} \right)^{\theta^j} \right)^{-1/\theta^j}$$

The constant Γ^{nj} depends on the trade elasticity θ^j and the within-sector CES elasticity η^{nj} (§2), and is well-defined provided $1 + \theta^j > \eta^{nj}$.

4.4 Trade shares (gravity equation)

$$\pi_t^{nj,ij} = \frac{\left(w_t^i \kappa_t^{nj,ij}\right)^{-\theta^j} \left(A_t^{ij}\right)^{\theta^j}}{\sum_{m=1}^N \left(w_t^m \kappa_t^{nj,mj}\right)^{-\theta^j} \left(A_t^{mj}\right)^{\theta^j}} \text{ (orig. eq. simplified 7)} \quad (5)$$

This is textbook multi-region, multi-sector Eaton-Kortum — sectors no longer linked to each other via an input-output network.

4.5 Goods-market clearing (household consumption expenditure only)

With no rentiers (no portfolio remittance term $\iota^n \chi_t$) and no materials (no input-output term), total expenditure on sector- j goods in region n is **purely household final consumption**:

$$X_t^{nj} = \alpha^j \sum_{k=1}^J w_t^{nk} L_t^{nk} \text{ (orig. eq. simplified 8)} \quad (6)$$

4.6 Labor-market clearing

$$w_t^{nj} L_t^{nj} = \sum_{i=1}^N \pi_t^{ij,nj} X_t^{ij} \text{ (orig. eq. simplified 9)} \quad (7)$$

Total revenue earned by region- n , sector- j firms — summed across every region i that buys from them — equals the wage bill. (Orig. eq. 10, structures market clearing, is dropped entirely: there is no structures factor to clear a market for.)

5 Equilibrium definitions (adapted)

Temporary equilibrium (orig. Definition 1, simplified): given (L_t, A_t, κ_t) , a temporary equilibrium is wages $w_t = \{w_t^{nj}\}$ together with $\pi_t = \{\pi_t^{nj,ij}\}$ and $X_t = \{X_t^{nj}\}$ solving §4.3–4.6 — $N \times J$ equations in $N \times J$ wages, given labor supply and fundamentals. This block is *static* in the sense that matters: it references no other time period, no expectation, no β — even though its solution changes every period because L_t does.

Sequential competitive equilibrium (orig. Definition 2, simplified): given $(L_0, \{A_t, \kappa_t\}_{t=0}^\infty, \Upsilon, b)$, a sequence $\{L_t, \mu_t, V_t, w_t\}_{t=0}^\infty$ solving the dynamic block (§3.2–3.4) and the temporary equilibrium (§4) at every t .

Stationary equilibrium (orig. Definition 3): a sequential competitive equilibrium in which $\{L_t, \mu_t, V_t, w_t\}$ are constant over time — fundamentals must also be constant, though households still move between markets each period as long as inflows equal outflows.

6 Dynamic hat algebra, simplified

Denote by $\dot{y}_{t+1} \equiv y_{t+1}/y_t$ the proportional (time-difference) change in any variable between periods t and $t+1$. The method's payoff is that the system below solves for the entire transition path without

ever needing the *level* of any fundamental — only the initial allocation (L_t, π_t, X_t) and the change in fundamentals $\dot{\Theta}_{t+1} = (\dot{A}_{t+1}, \dot{\kappa}_{t+1})$.

6.1 Temporary equilibrium in time differences (simplified orig. Proposition 1, eqs. 11–15)

Applying $\xi^n = 0$, $\gamma^{nj} = 1$, no input-output terms, and no rentiers to the original system:

$$\dot{x}_{t+1}^{nj} = \dot{w}_{t+1}^{nj} \quad (8)$$

$$\dot{P}_{t+1}^{nj} = \left(\sum_{i=1}^N \pi_t^{nj,ij} \left(\dot{w}_{t+1}^{ij} \dot{\kappa}_{t+1}^{nj,ij} \right)^{-\theta^j} \left(\dot{A}_{t+1}^{ij} \right)^{\theta^j} \right)^{-1/\theta^j} \quad (9)$$

$$\pi_{t+1}^{nj,ij} = \pi_t^{nj,ij} \left(\frac{\dot{w}_{t+1}^{ij} \dot{\kappa}_{t+1}^{nj,ij}}{\dot{P}_{t+1}^{nj}} \right)^{-\theta^j} \left(\dot{A}_{t+1}^{ij} \right)^{\theta^j} \quad (10)$$

$$X_{t+1}^{nj} = \alpha^j \sum_{k=1}^J \dot{w}_{t+1}^{nk} \dot{L}_{t+1}^{nk} w_t^{nk} L_t^{nk} \quad (11)$$

$$\dot{w}_{t+1}^{nj} \dot{L}_{t+1}^{nj} w_t^{nj} L_t^{nj} = \sum_{i=1}^N \pi_{t+1}^{ij,nj} X_{t+1}^{ij} \quad (12)$$

Equation (11)'s unit-cost change collapses to just the wage change because $\xi^n = 0$ kills the structures term and $\gamma^{nj} = 1$ removes its own exponent; the price-index and trade-share changes (12)–(13) carry A at the plain exponent θ^j rather than $\theta^j \gamma^{ij}$; the expenditure change (14) loses both the input-output term and the rentier term $\iota^n \chi_{t+1}$; and the labor-market-clearing change (15) loses its $\gamma^{nj}(1 - \xi^n)$ prefactor. Notably, χ_{t+1} does not need to be dropped by hand — its own formula carries a factor of $\xi^i/(1 - \xi^i)$ that is already zero for every region once structures are gone, so it vanishes on its own.

6.2 Sequential equilibrium in time differences (orig. Proposition 2, eqs. 16–18) — unchanged

This system is pure household block, untouched by the production-side simplification, since it never depended on structures or materials in the first place:

$$\mu_{t+1}^{nj,ik} = \frac{\mu_t^{nj,ik} (\dot{u}_{t+2}^{ik})^{\beta/\nu}}{\sum_{m=1}^N \sum_{h=0}^J \mu_t^{nj,mh} (\dot{u}_{t+2}^{mh})^{\beta/\nu}} \quad (13)$$

$$\dot{u}_{t+1}^{nj} = \dot{\omega}^{nj} (\dot{L}_{t+1}, \dot{\Theta}_{t+1}) \left(\sum_{i=1}^N \sum_{k=0}^J \mu_t^{nj,ik} (\dot{u}_{t+2}^{ik})^{\beta/\nu} \right)^\nu \quad (14)$$

$$L_{t+1}^{nj} = \sum_{i=1}^N \sum_{k=0}^J \mu_t^{ik,nj} L_t^{ik} \quad (15)$$

The connective piece linking the two systems above: $\dot{\omega}_{t+1}^{nj} = \dot{w}_{t+1}^{nj} / \dot{P}_{t+1}^n$, where $\dot{P}_{t+1}^n = \prod_{k=1}^J \left(\dot{P}_{t+1}^{nk} \right)^{\alpha^k}$ — the region-level price index change, following directly from the Cobb-Douglas aggregator (the α^k normalizing constants cancel in the ratio). For the non-employment state, $\dot{\omega}_{t+1}^{n0} = 1$ always, since b^n never changes.

7 Solving for the transition path: the Appendix D algorithm

Given $L_0, \pi_0, X_0, \mu_{-1}$, and a path of fundamental changes $\dot{\Theta}_t = (\dot{A}_t, \dot{\kappa}_t)$ for $t = 1, 2, \dots$ (in practice truncated at a large horizon T , exploiting the convergence property $\dot{\Theta}_t \rightarrow 1$), the transition path of the sequential competitive equilibrium can be computed by a guess-forward-simulate-backward-solve fixed-point procedure (orig. Appendix D). Note X_0 is itself redundant, recoverable from π_0 and the wage bill implied by L_0 and production; and w_0 need not be listed separately, since that information is already carried by X_0/π_0 . By contrast μ_{-1} — last period’s realized migration shares — cannot be recovered from the stocks L_{-1}, L_0 alone (gross flows are not identified from net stocks) and so must be supplied as data, seeding the μ recursion below.

1. **Guess** a full path $\{\dot{u}_t\}_{t=0}^T$, with terminal condition $\dot{u}_{T+1} = 1$.
2. **Forward-simulate migration shares** $\{\mu_t\}_{t=0}^T$ via eq. (16), seeded by μ_{-1} and the guessed path — each μ_t is built from μ_{t-1} and $\dot{u}_{t+1}$.
3. **Forward-simulate labor** $\{L_t\}_{t=1}^{T+1}$ via eq. (18). The simulation runs one period past T because μ_T generates L_{T+1} , which the next step requires.
4. **Solve the temporary equilibrium** period by period, $t = 0, \dots, T$: form $\dot{L}_{t+1}$ from $\{L_t\}$, guess $\dot{w}_{t+1}^{nj}$, use it to obtain $\dot{x}_{t+1}^{nj}, \dot{P}_{t+1}^{nj}, \pi_{t+1}^{nj,ij}$, and X_{t+1}^{nj} via eqs. (11)–(14), and check that labor-market clearing, eq. (15), holds; revise the wage guess and repeat until it does. The wage-bill level $w_t^{nk} L_t^{nk}$ must be tracked forward across iterations — not just the growth rate $\dot{w}$ — since X_{t+1}^{nj} depends on the compounding wage-bill level itself, not merely its rate of change.
5. **Backward-solve** an updated path $\{\dot{u}_t\}_{t=0}^T$ via eq. (17): $\dot{u}_{t+1}^{nj}$ is built from the lagged share $\mu_t^{nj,ik}$, the real-wage change $\dot{\omega}_{t+1}^{nj} = \dot{w}_{t+1}^{nj} / \dot{P}_{t+1}^n$ just computed in step 4, and the next period’s $\dot{u}_{t+2}^{nk}$ — working backward from the terminal condition $\dot{u}_{T+1} = 1$ (the same value fixed in step 1, not a new assumption) down to $\dot{u}_0$.
6. **Compare** the updated path to the guess from step 1, via a sup-norm criterion: $\max_t \left| \dot{u}_t^{(1)} - \dot{u}_t^{(0)} \right| < \varepsilon$ for small ε .
7. **Iterate**: if the tolerance is not met, set the updated path as the new guess and repeat from step 2 until convergence.

Once $\{\dot{u}_t\}$ converges, the paths $\{L_t\}$, $\{\dot{w}_t\}$, and $\{\mu_t\}$ generated along the way constitute the full transition path of the sequential competitive equilibrium.

7.1 Numeraire, initial guesses, and a redundancy check

The initial guess for both loops can simply be “no change”: $\dot{u}_t = 1$ for all t , and $\dot{w}_{t+1}^{nj} = 1$ for all (n, j, t) in step 4. This matches the terminal condition $\dot{u}_{T+1} = 1$ and tends to be close to the fixed point provided $\dot{\Theta}_t$ is not too large a shock.

Eq. (15) is homogeneous of degree zero in $\dot{w}_{t+1}$ — scaling every market’s wage guess by a common factor leaves both sides unchanged in ratio — so it pins down only *relative* wages within a period,

not levels of change. Step 4 therefore needs a numeraire; take region 1, sector 1 as the reference market, $\dot{w}_{t+1}^{1,1} \equiv 1$ for every t , leaving $NJ - 1$ free unknowns.

This also exposes a redundancy worth checking numerically. Since $\pi_{t+1}^{ij,nj}$ is a genuine trade share at $t+1$ (built from $\pi_t^{ij,nj}$ in eq. (13)), it sums to one across source regions for any fixed buyer market: $\sum_{n=1}^N \pi_{t+1}^{ij,nj} = 1$. Summing eq. (15) over every market (n, j) :

$$\sum_{n,j} \dot{w}_{t+1}^{nj} \dot{L}_{t+1}^{nj} w_t^{nj} L_t^{nj} = \sum_{i,j} X_{t+1}^{ij} \underbrace{\sum_n \pi_{t+1}^{ij,nj}}_{=1} = \sum_{i,j} X_{t+1}^{ij},$$

and by eq. (14), $\sum_j X_{t+1}^{ij} = \sum_j \alpha^j \sum_k \dot{w}_{t+1}^{ik} \dot{L}_{t+1}^{ik} w_t^{ik} L_t^{ik} = \sum_k \dot{w}_{t+1}^{ik} \dot{L}_{t+1}^{ik} w_t^{ik} L_t^{ik}$ (since $\sum_j \alpha^j = 1$) — exactly the wage bill summed over region i . So both sides of eq. (15), summed over *all* markets, are identical for *any* wage vector, equilibrium or not. Only $NJ - 1$ of the NJ instances of eq. (15) are therefore independent: solve any $NJ - 1$ of them for the $NJ - 1$ non-numeraire wages, and the remaining equation holds automatically — evaluating its residual at the solution is a useful numerical check on the solve.

8 What changed vs. the original paper, at a glance

Object	Original	Simplified
Production inputs	labor, structures, materials	labor only
Constant fundamentals $\bar{\Theta}$	(Y, H, b)	(Y, b)
A^{ij} exponent (price index / trade share)	$\theta^j \gamma^{ij}$	θ^j
X_t^{nj} (sector- j expenditure)	materials demand + final consumption	final consumption only
Structures market clearing (orig. eq. 10)	present	dropped
Rentier/portfolio ($\iota^n \chi_t$)	present	dropped
Household dynamic block (§3)	—	unchanged

Status: this reflects simplification #1 only. The household block, equilibrium structure, dynamic hat algebra (Propositions 1–2, orig. Section 3), and the Appendix D solution algorithm are now reformulated above. Proposition 3 (counterfactuals relative to a baseline) has not yet been written out for the simplified model.